

Lexie Linear Algebra Homework

Chapter 5 - Index, Slice, and Reshape NumPy Arrays

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Purpose: practice the mechanics of working with NumPy arrays as data tables. You should be able to inspect shape, grab rows and columns, split a table into input features X and output target y, and reshape y when a later algorithm expects a column vector.

Part A - Straightforward NumPy Practice

Copy or use the starter file. Run your code often. Print both the array values and the shapes.

```
import numpy as np
```

```
data = np.array([
    [10, 2, 100],
    [20, 4, 200],
    [30, 6, 300],
    [40, 8, 400],
    [50, 10, 500],
    [60, 12, 600]
])
```

- Print the shape of data.
- Print the first row.
- Print the last row using negative indexing.
- Print the first column.
- Print the last column.
- Split data into X and y, where X contains the first two columns and y contains the last column.
- Print X.shape and y.shape. Explain in one sentence what each dimension means.
- Reshape y into a column vector called y_col and print y_col.shape.
- Split the first 4 rows into train and the last 2 rows into test.

Part B - Practical Problem: Supervised Learning Table

This table represents a tiny supervised-learning dataset. Each row is one student. The first two columns are input features. The last column is the target label.

```
students = np.array([
    [2, 75, 0],
    [4, 82, 1],
    [1, 60, 0],
    [5, 90, 1],
    [3, 78, 1],
    [2, 65, 0],
    [6, 92, 1],
    [1, 58, 0]
])
```

```
# Each row means:  
# [hours_studied, practice_score, passed_final]
```

- Split the table into X and y.
- Print the number of students using shape.
- Print the number of input features using shape.
- Make y into a column vector called y_col.
- Create train_X, train_y, test_X, and test_y using the first 6 students for training and the last 2 for testing.
- Write a short comment explaining why X is 2D but y often starts as 1D.

Reflection Questions

- What does data[:, :-1] mean in plain English?
- What does data[:, -1] mean in plain English?
- Why is y.reshape((y.shape[0], 1)) useful?